TASK SPECIFICATION: P4-05 PHASE P4 — FINANCIAL LEDGER & TUITION OFFSET ENGINE

Tuition Payment & Digital Receipt UI

Git Branch: `feature/P4-05-tuition-ui` · Target Pull Request: feature/P4-05-tuition-ui

Core Objective & Technical Scope:
Tuition fee payment interface, credit offset modal, and printable official institutional payment receipt with validation QR code.

Priority Level	Medium	Complexity Rating	`M`
Execution Status	Backlog 📋	Target PR / Commit	`feature/P4-05-tuition-ui`
Upstream Dependencies	P4-03, P4-04		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Build TuitionOverview displaying current term fees and pending balance.
2. Implement payment modal allowing partial or full offset from wallet earnings.
3. Build printable official institutional receipt view with formatted print CSS.
4. Test complete payment flow end-to-end.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `frontend/src/features/tuition/*`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Accurate balance calculation preventing invalid payment attempts.
- Receipt prints cleanly on A4 with university header and validation code.

VERIFICATION & TEST COVERAGE STRATEGY

- TuitionForm validation tests
- Receipt print CSS test
- Playwright payment flow